

CINEWAVE ENTERTAINMENT

MOVIE TICKET BOOKING MANAGEMENT SYSTEM

Pega Blueprint Project Report

CINEWAVE — MOVIE TICKET REQUEST LIFECYCLE



Customer request → seat/capacity verification → payment decision → ticket generation → case completion

Application: Ticketing and Booking
Case Type: Movie Ticket Request
Business: CineWave Entertainment
Prepared by: Kolla Aditya

CERTIFICATE

This is to certify that the project titled "Movie Ticket Booking Management System" has been documented as a Pega Blueprint case-management solution for CineWave Entertainment. The report describes the configured business workflow, data model, validations, routing, payment handling, SLA and test evidence captured during project development.

DECLARATION

I declare that this project documentation represents the Movie Ticket Request solution developed and tested during the project work. The screenshots included in this report are project evidence captured from the Pega Blueprint environment and preview flow.

ACKNOWLEDGEMENT

I sincerely thank the faculty, trainers and mentors who provided guidance during the development of this Pega Blueprint project. I also acknowledge the Pega Blueprint environment and the project evaluation process that helped structure the movie ticket booking requirements into a case-management workflow.

ABSTRACT

The Movie Ticket Booking Management System is a workflow-driven application modelled using Pega Blueprint for CineWave Entertainment. The system represents a movie booking request as a case that progresses through Request Details, Availability, Approval, Booking Execution and Resolved. The solution captures movie/show information, ticket quantity, seat information, pricing, payment, status, routing and SLA. Actual Pega screenshots are included, and the runtime seat-picker limitation is documented transparently.

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1. Project Overview

CineWave Entertainment uses a Movie Ticket Request case within the Ticketing and Booking application. The configured lifecycle is Request Details → Availability → Approval → Booking Execution → Resolved.

2. Business Problem

The process must keep ticket quantity, seats, capacity, pricing, payment, cancellation and final booking information consistent.

3. Project Objectives

The project models the complete request, validates capacity and seat count, calculates total cost, restricts payment, supports confirmation/cancellation, routes shows, tracks SLA and generates ticket information.

4. Existing and Proposed Process

The proposed case process replaces scattered booking activities with controlled stages, validation, routing, status and SLA tracking.

5. Technology and Platform

Pega Blueprint and Pega case-management concepts are used for workflow, data, rules, personas, routing, SLA and preview validation.

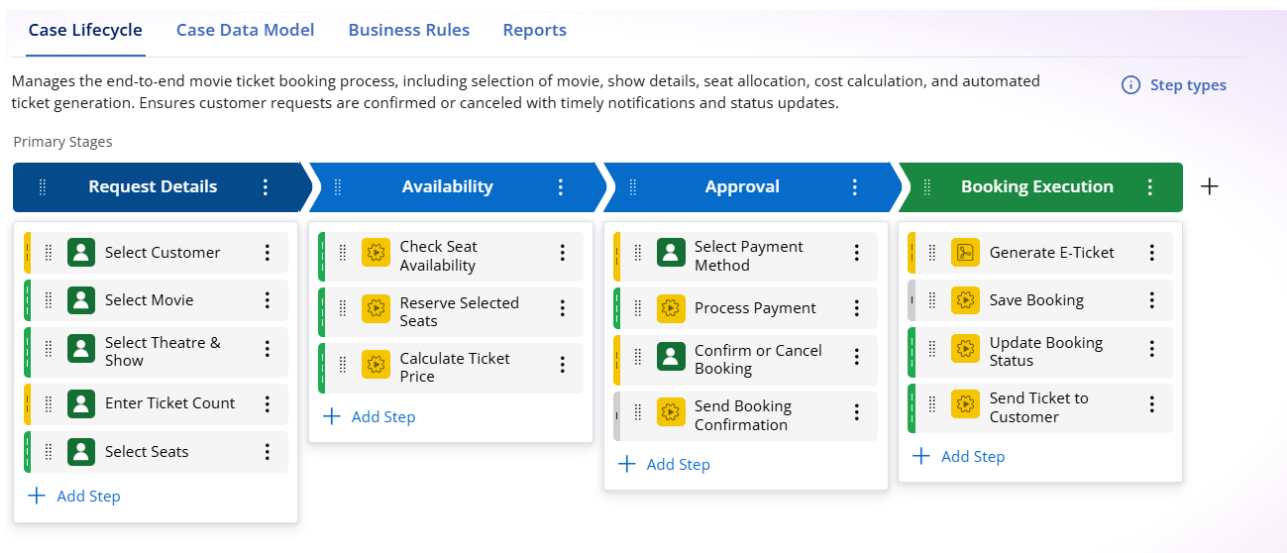


Figure 1. Actual Pega workflow screenshot.

6. Solution Architecture

CINEWAVE — DATA ARCHITECTURE

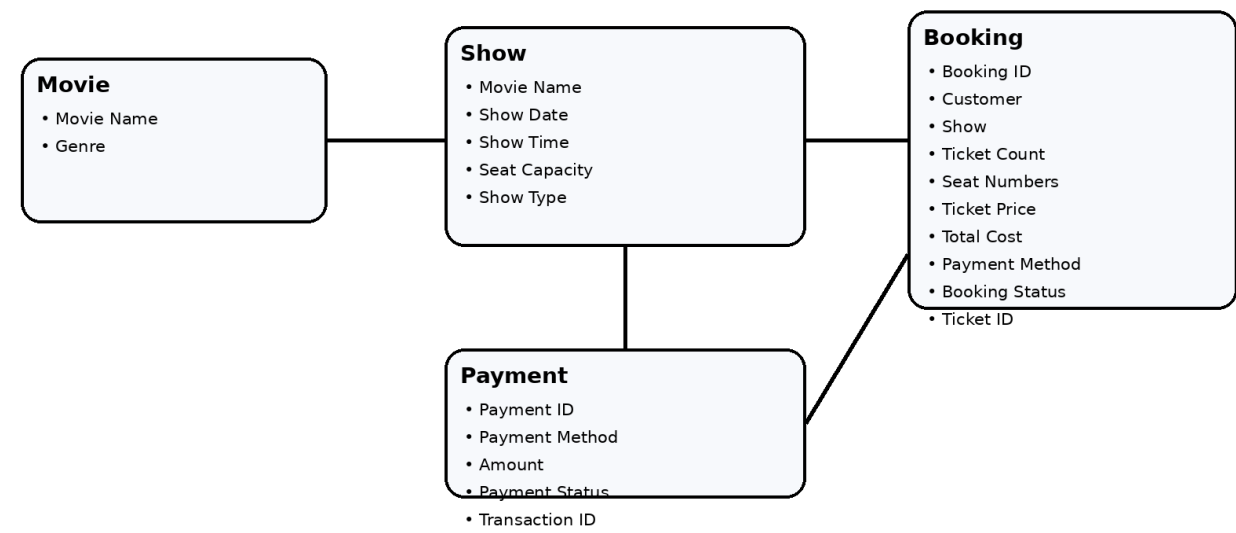


Figure 2. Logical data architecture.

7. Case Lifecycle

CINEWAVE — MOVIE TICKET REQUEST LIFECYCLE



Customer request → seat/capacity verification → payment decision → ticket generation → case completion

Figure 3. Five-stage lifecycle.

8. Detailed Stage Design

Steps	
Select Customer; Select Movie; Select Theatre & Show; Enter Ticket Count; Select Seats	Customer-facing request information.
Check Seat Availability; Reserve Selected Seats; Calculate Ticket Price	Capacity, seat reservation and pricing.
Select Payment Method; Process Payment; Confirm or Cancel Booking; Send Booking Confirmation	Payment and customer decision.
Generate E-Ticket; Save Booking; Update Booking Status; Send Ticket to Customer	Finalize booking and communicate ticket.
Successful completion of the case	Final lifecycle state.

Manages the end-to-end movie ticket booking process, including selection of movie, show details, seat allocation, cost calculation, and automated ticket generation. Ensures customer requests are confirmed or canceled with timely notifications and status updates.

[Step types](#)

Primary Stages

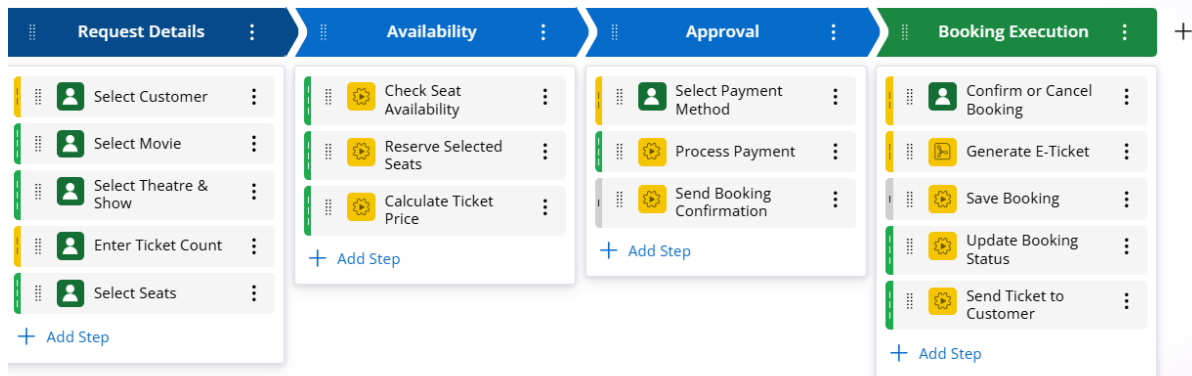


Figure 4. Actual workflow configuration.

9. Field Specification

Field	Type / Value	Purpose
Movie Name	Movie / text	Selected movie.
Show Date	Date	Show date.
Show Time	Time	Show time.
Number of Tickets	Integer	Quantity.
Show Type	Premium / Standard	Routing.
Seat Allocation	Seat information	Selected seats.
Available Seats Count	Integer	>= tickets.
Ticket Price	Currency	Price per ticket.
Total Booking Cost	Currency	Price × quantity.
Payment Method	UPI / Credit-Debit Card	Payment choice.
Booking Status	Pending / Confirmed / Cancelled	Booking state.
Cancellation Reason	Text	Required if cancelled.
Ticket ID	Text	Ticket identifier.
Execution Status	Status	Execution tracking.

Case Data Model

[Add a Field](#)

[Generate More](#)

Field Name	Description	Type	Primary ⓘ
Movie ticket requ...	The canonical name for this movie ticket request, representing the main identifier f...	Text	✓ ...
Movie name	Name of the selected movie for which the ticket is being requested.	Text	✓ ...
Show date	Date of the movie show for which the booking is being made.	Date	✓ ...
Show time	Time of the movie show for which the booking is being made.	Time	✓ ...
Number of tickets	Total number of tickets requested by the customer for the selected show.	Integer	✓ ...
Show type	Indicates whether the show is premium or standard for routing purposes and SLA ...	Text	...
Booking status	Current booking status supporting Pending, Confirmed, Cancelled only.	Text	...
Total booking cost	Total calculated cost for the booking, based on ticket quantity and selected show ty...	Currency	...
Seat allocation	Allocated seat(s) for the requested tickets after confirmation.	Text	...
SLA timer	Timer tracking service level agreement for timely booking processing and notificatio...	DateTime	...

Figure 5. Actual case data/model evidence.

10. Business Rules and Validations

Rule	Logic	Result
Availability	Available >= tickets	Proceed
Insufficient	Available < tickets	Block
Seat count	Selected = tickets	Block mismatch
Pricing	Price × tickets	Calculate total
Payment	UPI / Credit-Debit	Allow
Cancellation	Cancelled requires reason	Require reason
Routing	Premium / Standard	Queue routing

11. Data Architecture

CINEWAVE — DATA ARCHITECTURE

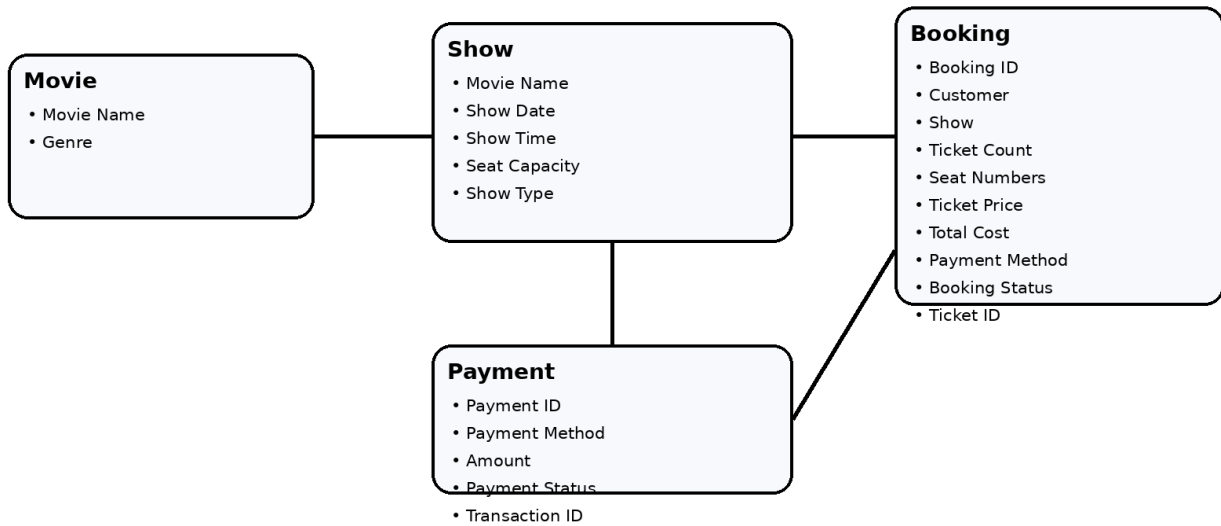


Figure 6. Data object architecture.

12. Personas and Access

CINEWAVE — PERSONA & RESPONSIBILITY MODEL

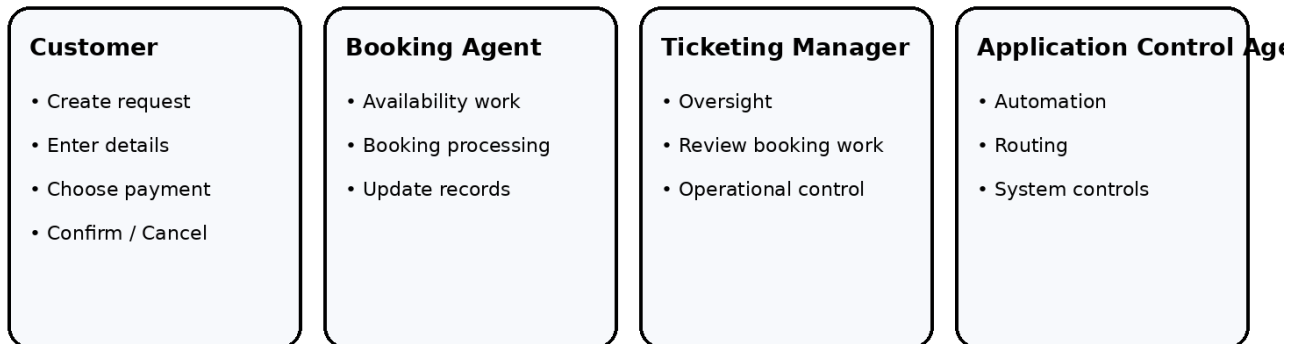
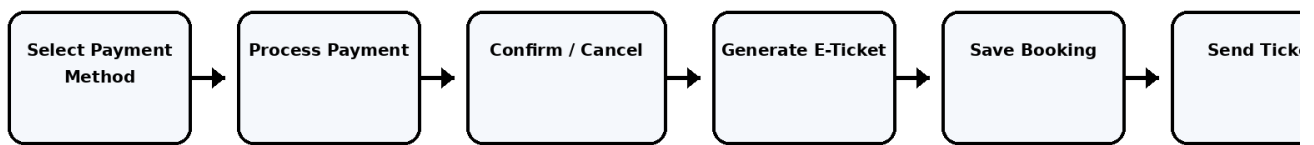


Figure 7. Persona model.

13. Payment Management

CINEWAVE — PAYMENT & BOOKING FLOW



Allowed payment choices: UPI • Credit/Debit Card

Figure 8. Payment flow.

How is this field captured? *

☐ User input
The user types in information using a form.

☐ Calculated
A read-only value is automatically calculated using business logic

☒ Picklist
The user selects an option from a picklist.

Picklist options *

⋮

UPI

⋮

⋮

Credit/Debit Card

⋮

[+ Add Option](#)

Cancel

Save

Figure 9. Actual Payment Method screenshot.

14. Premium / Standard Routing

Show Type	Queue
Premium	PremiumShowQueue
Standard	StandardShowQueue

15. SLA Management

Item	Value
Goal	1 day
Deadline	2 days
Approaching deadline	Flag / priority handling

16. Booking Execution

Generate E-Ticket → Ticket ID → Save Booking → Update Booking Status/Execution Status → Send Ticket → Resolved.

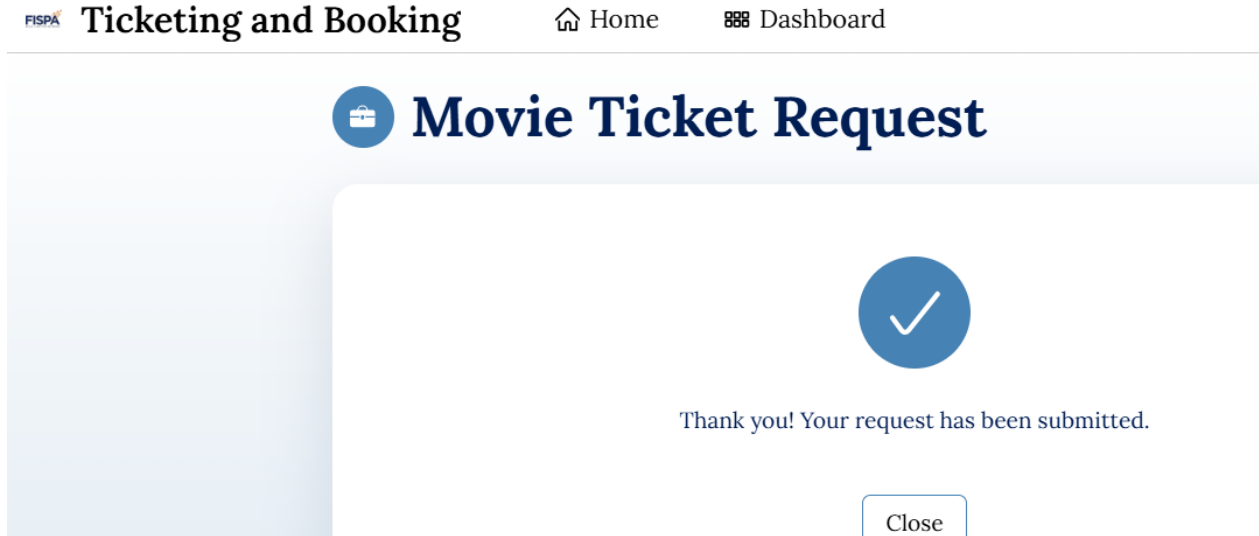


Figure 10. Actual successful preview submission.

17. Testing Strategy

Test	Expected Result
2 tickets + 2 seats	Continue
5 tickets + 5 seats	Continue
5 tickets + 2 seats	Block
Insufficient capacity	Block
UPI	Allowed
Credit/Debit Card	Allowed
Cancelled without reason	Block
Cancelled with reason	Continue

18. Actual Pega Screenshots

Blueprint recommendations



Best Practices 2

We have found opportunities to improve your Blueprint through our Best Practices.

No access settings

PERSONAS **BEST PRACTICE**

- Personas affected: Customer, Booking Agent, Ticketing Manager
- Go to [Personas](#)

No referenced External SOR

DATA & INTEGRATIONS **BEST PRACTICE**

- Data & Integrations affected: All instances
- Go to [Data & Integrations](#)

Close

Download anyway

Actual uploaded Pega screenshot — 2707e306-e1d3-45d0-bd0f-38d47ef4f461.png

Case Data Model

Add a Field

Generate More

Field Name	Description	Type	Primary
Movie ticket reque...	The canonical name for this movie ticket request, representing the main identifier f...	Text	☒ ...
Movie name	Name of the selected movie for which the ticket is being requested.	Text	☒ ...
Show date	Date of the movie show for which the booking is being made.	Date	☒ ...
Show time	Time of the movie show for which the booking is being made.	Time	☒ ...
Number of tickets	Total number of tickets requested by the customer for the selected show.	Integer	☒ ...
Show type	Indicates whether the show is premium or standard for routing purposes and SLA ...	Text	☐ ...
Booking status	Current booking status supporting Pending, Confirmed, Cancelled only.	Text	☐ ...
Total booking cost	Total calculated cost for the booking, based on ticket quantity and selected show ty...	Currency	☐ ...
Seat allocation	Allocated seat(s) for the requested tickets after confirmation.	Text	☐ ...
SLA timer	Timer tracking service level agreement for timely booking processing and notificatio...	DateTime	☐ ...

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Case Lifecycle

Case Data Model

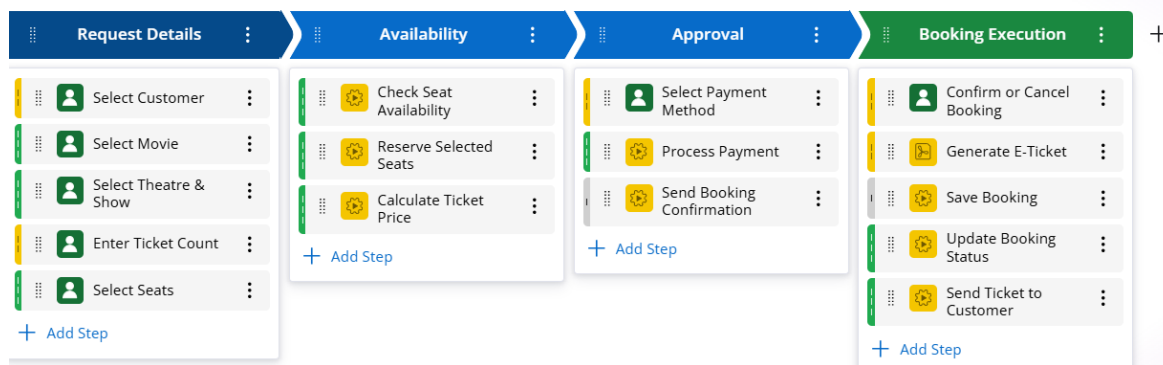
Business Rules

Reports

Manages the end-to-end movie ticket booking process, including selection of movie, show details, seat allocation, cost calculation, and automated ticket generation. Ensures customer requests are confirmed or canceled with timely notifications and status updates.

Step types

Primary Stages



Actual uploaded Pega screenshot — b570baa0-9613-4f1a-b80b-3fcc2327e5dc.png

FISPA

Ticketing and Booking

Home

Dashboard

A

Movie Ticket Request

Booking Request

Select Movie

Movie name

Crimson Tides

Show date

09 / 22 / 2026

Show time

06:30 PM

Show type

Premium

Number of tickets

5

Seat allocation

H11, H12

Cancel

Submit

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PRIVACY

COMMUNITY

ABOUT

Actual uploaded Pega screenshot — 3b4e5f99-8a8e-4cd5-933d-cbd1e94e2853.png

FISPA

Ticketing and Booking

Home

Dashboard

Movie Ticket Request

Booking Request

Select Customer

Customer contact email

lisa.kumar@email.com

Cancel

Submit

Actual uploaded Pega screenshot — e3291218-d09e-4d8c-810e-4e19eeb5fab1.png

FISPA

Ticketing and Booking

Home

Dashboard

Movie Ticket Request

Thank you! Your request has been submitted.

Close

Actual uploaded Pega screenshot — 60e636ed-05bb-4487-bb01-22fd9d6aaa4f.png

Submit

How is this field captured? *

The user selects an option from a picklist.

Picklist options *

[+ Add Option](#)

Save

Product Design

Edit Field: **Payment Method**



Field Name *

Payment Method

Field Type *

Text

Description *

Specifies the payment option selected by the customer, such as UPI, Credit/Debit Card, or Net Banking.

☐ Mark as Primary Field

How is this field captured? *

☐ User input ☐ Calculated ☒ Picklist

Cancel

Save

Actual uploaded Pega screenshot — 1b6bfe0c-a42f-4c8c-a6a8-a7a521e23891.png

☐ Mark as Primary Field

How is this field captured? *

☐ User input

The user types in information using a form.

☐ Calculated

A read-only value is automatically calculated using business logic

☒ Picklist

The user selects an option from a picklist.

Picklist options *



UPI



Credit/Debit Card



Net Banking



Cancel

Save

Actual uploaded Pega screenshot — 7d295354-37a1-4b59-8981-497df7cb1d58.png

AI Assistant

Define workflow details for: Movie Ticket Request

Case Lifecycle

Case Data Model

Business Rules

Reports

Manages the end-to-end movie ticket booking process, including selection of movie, show details, seat allocation, cost calculation, and automated ticket generation canceled with timely notifications and status updates.

Primary Stages

Booking Request

Select Customer

Select Movie

Select Theatre & Show

Enter Ticket Count

Select Seats

Add Step

Availability

Check Seat Availability

Reserve Selected Seats

Calculate Ticket Price

Add Step

Payment & Confirm...

Select Payment Method

Process Payment

Send Booking Confirmation

Add Step

Ticket Generation

Confirm or Cancel Booking

Generate E-Ticket

Save Booking

Update Booking Status

Send Ticket to Customer

Add Step

Actual uploaded Pega screenshot — 3a228e24-6aed-41e0-bff9-e79d22f5e06f.png

Field Configuration - Field

[Back to Select Payment Method](#)

Visibility

Always

Required

Never

Read-only

Never

Cancel

Save

Actual uploaded Pega screenshot — 7e032ad9-59ab-4ccb-8121-5fa57bad8570.png

CineWave Entertainment | Movie Ticket Booking Management System | Pega Blueprint

Select Payment Method

Stage: Payment & Confirmation • Step type: Assignment

General

SLA

Type *

Collect information

Name *

Select Payment Method

Route To

Customer

☐ Enable agentic assignment processing

Fields

Field Name	Conditions
Total booking cost	
Ticket review status	
Payment Method	

Cancel

Save

Actual uploaded Pega screenshot — 3970aa84-cad1-4416-a887-ea505db377bb.png

Pega Sites

Hello Aditya

Define workflow details for: **Movie Ticket Request**

Case Lifecycle

Case Data Model

Business Rules

Reports

Manages the end-to-end movie ticket booking process, including selection of movie, show details, seat allocation, cost calculation, and automated ticket generation. Ensures customer requests are confirmed or canceled with timely notifications and status updates.

Step types

Primary Stages

Booking Request

Select Customer

Select Movie

Select Theatre & Show

Enter Ticket Count

Select Seats

Add Step

Availability

Check Seat Availability

Reserve Selected Seats

Calculate Ticket Price

Add Step

Payment & Confirm...

Select Payment Method

Process Payment

Send Booking Confirmation

Add Step

Ticket Generation

Confirm or Cancel Booking

Generate E-Ticket

Save Booking

Update Booking Status

Send Ticket to Customer

Add Step

Back

Preview my app

Save & Close

Next

Actual uploaded Pega screenshot — 52e24e4f-d19f-4c4f-a9c4-f840dd7acc0.png

Field Configuration - Booking cancellation reason



[Back to Confirm or Cancel Booking](#)

Visibility

Always

▼

Required

Never

▼

Read-only

Never

▼

Cancel

Save

Actual uploaded Pega screenshot — 5cf01187-49c8-439c-8ca7-c22908a0dd33.png

Field Configuration - Booking status



[Back to Confirm or Cancel Booking](#)

Visibility

Always

▼

Required

Always

▼

Read-only

Never

▼

Actual uploaded Pega screenshot — 157b4334-42a2-4e26-96df-e397bd4219fa.png

Field Configuration - Booking status



[Back to Confirm or Cancel Booking](#)

Visibility

Always

▼

Required

Never

▼

Read-only

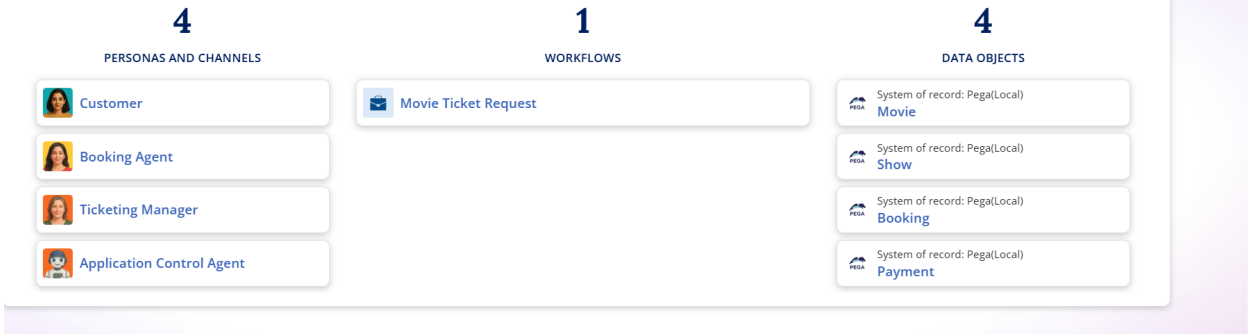
Never

▼

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Application Architecture

Users access the application through defined channels, assume specific roles, and trigger workflows that manage how work progresses. Those workflows operate on shared Data Objects to ensure the system stays consistent and up to date.



Actual uploaded Pega screenshot — d2682b0e-697a-4b40-9d1e-db63be97671d.png

This screenshot shows the "Case Lifecycle" for a "Movie Ticket Request". The interface includes a sidebar with "AI Assistant" and a top navigation bar with "Case Lifecycle", "Case Data Model", "Business Rules", and "Reports". The main content area describes the end-to-end movie ticket booking process. Below the description, the "Primary Stages" are listed:

- Booking Request:** Select Customer, Select Movie, Select Theatre & Show, Enter Ticket Count, Select Seats.
- Availability:** Check Seat Availability, Reserve Selected Seats, Calculate Ticket Price.
- Payment & Confirm...:** Select Payment Method, Process Payment, Send Booking Confirmation.
- Ticket Generation:** Confirm or Cancel Booking, Generate E-Ticket, Save Booking, Update Booking Status, Send Ticket to Customer.

Buttons for "Back" and "Preview my app" are visible at the bottom.

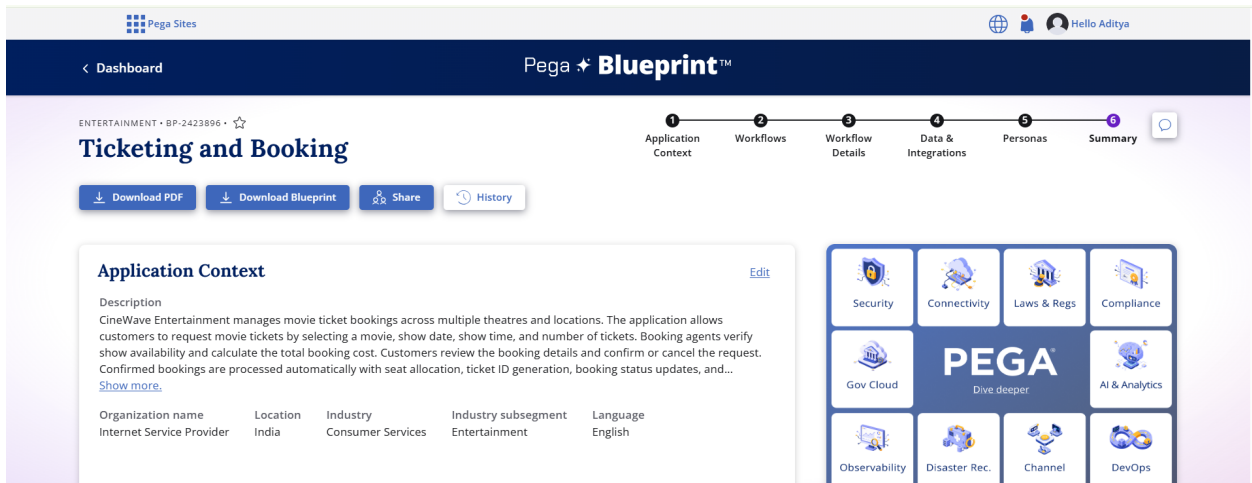
Actual uploaded Pega screenshot — 8271e3cc-1b1b-4188-96a4-2d1e6cdf5d3.png

This screenshot shows the "Case Lifecycle" for a "Movie Ticket Request" with more detailed steps. The interface includes a sidebar with "AI Assistant" and a top navigation bar with "Application Context", "Workflows", "Workflow Details", "Data & Integrations", "Personas", and "Summary". The main content area describes the end-to-end movie ticket booking process. Below the description, the "Primary Stages" are listed:

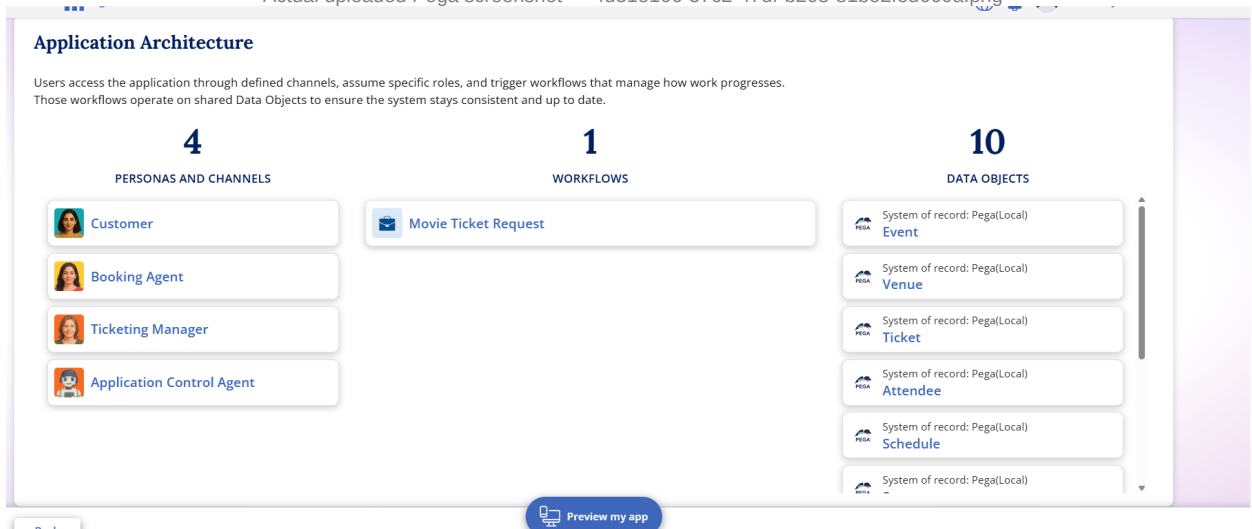
- Booking Request:** Select Customer, Select Movie, Select Theatre & Show, Enter Ticket Count, Select Seats.
- Availability:** Check Seat Availability, Reserve Selected Seats, Calculate Ticket Price, Validate Wallet Balance.
- Payment & Confirm...:** Select Payment Method, Process Payment, Reduce Wallet Balance, Send Booking Confirmation.
- Ticket Generation:** Generate E-Ticket, Save Booking, Update Booking Status, Send Ticket to Customer.

Buttons for "Back", "Preview my app", "Save & Close", and "Next" are visible at the bottom.

Actual uploaded Pega screenshot — 47cb608b-8e56-4d1f-8163-87313a0e72c1.png



Actual uploaded Pega screenshot — 4d31e166-57c2-47df-b208-81be2fed660a.png



Actual uploaded Pega screenshot — ea0855a5-045b-43aa-bbe7-ffb8a633670f.png

19. Runtime Observation and Limitation

Preview testing identified a seat-allocation mismatch where a larger Number of Tickets could still result in fewer seats being shown/allocated. The Blueprint configuration can define the validation that selected/allocated seats equal ticket quantity, but it does not directly control the low-level runtime seat picker needed to automatically allocate the next N available seats. This report therefore does not claim dynamic automatic seat allocation as completed.

20. Future Enhancements

Enhancement	Purpose
Dynamic seat allocation	Allocate exactly N available seats.
Live inventory	Read real-time capacity.
Payment gateway	Process real transactions.
QR e-ticket	Scannable entry ticket.
History	Customer booking/cancellation history.
Analytics	Revenue, occupancy and SLA metrics.
Recovery	Handle failed execution.

21. Conclusion

The CineWave Movie Ticket Booking Management System demonstrates a structured Pega case-management solution with a defined lifecycle, reusable data objects, validations, payment controls, routing, personas, SLA and booking execution. Actual project screenshots are included, while the runtime seat-picker limitation is recorded honestly.